



FLSP: A federated learning method with self-adaptive privacy for ensuring high model performance in edge computing

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ABSTRACT

With the development of the Internet of Things (IoT), the enormous volumes of data generated by edge devices is both private and sensitive, making it unsuitable for centralized processing through traditional machine learning methods. Consequently, this paper proposes a synergistic approach that combines federated learning with edge computing to design a distributed scheme with self-adaptive privacy, called FLSP. To address the inherent limitations of federated learning in privacy preservation, this paper integrates differential privacy to enhance the security of data aggregation. Acknowledging that the direct application of differential privacy can significantly compromise model utility, as well as considering the challenges posed by heterogeneous client environments in edge computing, FLSP employs a simulated annealing algorithm alongside dynamic client model aggregation. This approach allows for the adaptive allocation of privacy budgets to individual clients, thereby improving security without significantly degrading model performance. In comparison to federated learning schemes that utilize differential privacy with a fixed privacy budget, experiments and security analyses demonstrate that the FLSP framework successfully protects privacy and meets security requirements while maintaining high model accuracy, even under conditions of unbalanced and non-independent and identically distributed (Non-IID) data.

1. Introduction

As Fifth-generation Network (5G) technology matures increasingly, the Internet of Things (IoT) technology, as the core driving force to promote the intelligent transformation of various industries, is reshaping our mode of production and life at an unprecedented speed. IoT facilitates interconnection and data sharing between devices by integrating massive physical devices and sensors into the network. This leads to a multitude of innovative application scenarios, including smart cities, industrial IoT, intelligent transportation, precision agriculture, which provide strong technical support for enhancing production efficiency, optimizing resource allocation, and improving quality of life. However, cloud computing is no longer sufficient to handle and analyze the vast volumes of data generated by IoT devices in real time, owing to the exponential growth in the number of IoT devices and the complexity of application scenarios. High latency, bandwidth bottlenecks, and data privacy and security risks arising from centralized processing of massive volumes of data in the cloud have increasingly become prominent, thus emerging as a key bottleneck restricting the further development of IoT.

In this context, edge computing has emerged, presenting new opportunities for IoT development. Centralized machine learning approaches rely on collecting training samples on centralized parameter servers. Transmitting massive amounts of data can cause significant delays. Furthermore, traditional centralized methods fail to guarantee user privacy, which is a core concern for numerous emerging applications such as virtual reality, autonomous driving, and drones. For these reasons, centralized machine learning is not an ideal choice. More importantly, limited communication resources often prevent edge devices from uploading data to servers for centralized processing. This necessitates distributed learning algorithms, enabling devices to collaboratively build unified models through local training.

Federated learning (FL) [1] features a distributed computing architecture, minimized data communication, model quality preservation, and data isolation. These properties make FL particularly well-suited for edge computing environments [2]. Fig. 1 illustrates an IoT architecture driven by edge computing and the application of FL. The figure presents a three-layer structure consisting of cloud server, edge server and edge device. The cloud server provides global resource scheduling and long-term storage support. The edge server acts as a key intermediate node,

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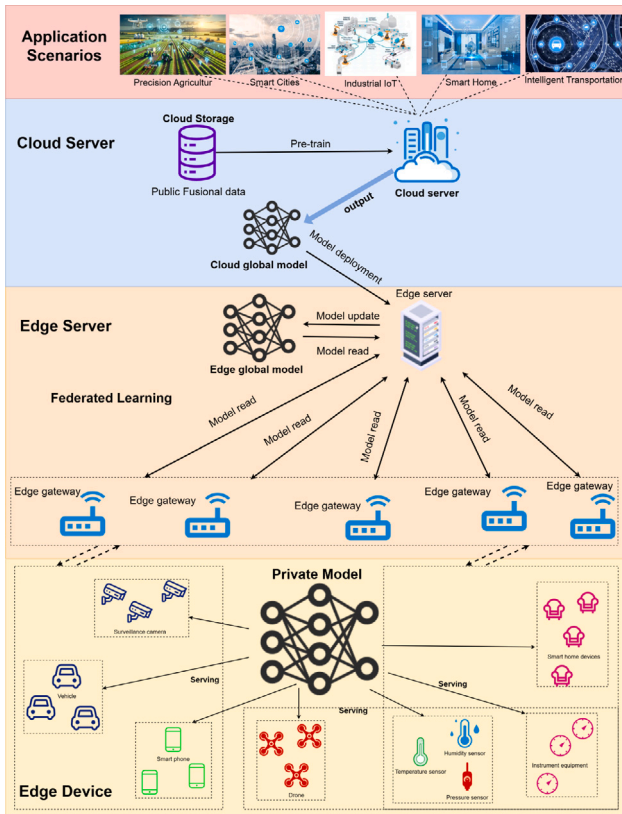


Fig. 1. System of edge computing driving IoT.

undertaking local model aggregation and federated learning tasks. Edge devices equipped with sensors generate massive private data and maintain local private models, conducting local training without uploading raw data. They only transmit model updates to the edge server, which aggregates to form an optimized global model and distributes it back for local fine-tuning. This architecture integrates edge computing's low-latency advantage with federated learning's data isolation feature, addressing key challenges of traditional cloud-centric IoT systems.

Moreover, FL based on edge computing addresses the data-driven requirements and privacy concerns faced by AI systems. Federated learning enables edge devices to collaborate on model training by sending only local model updates to the base station, while keeping local training data private. User privacy is protected because the data center does not have access to local datasets. FL enables distributed training across mobile devices, where updates from local computations are aggregated to construct shared models. However, data heterogeneity across diverse edge devices and privacy leakage risks during model update transmission still impose critical challenges to the model performance and privacy protection of federated learning.

Differential privacy (DP) is a privacy-preserving method widely applied in federated learning. By introducing perturbed noise into model updates, it ensures attackers cannot infer individual data values from the overall data distribution. However, the introduction of noise leads to a decline in model performance, a trade-off quantifiable by the privacy budget ϵ . The privacy budget ϵ serves as the core quantitative metric for evaluating an algorithm's privacy protection strength for individual user data. Its numerical value is inversely proportional to the level of privacy protection. A smaller ϵ necessitates injecting more noise into the local model update before uploading it to the server, thereby enhancing defense against inference attacks targeting sensitive user data. A larger ϵ reduces noise injection in model updates, helping preserve the validity of locally trained results and better maintaining overall model performance. Therefore, it is particularly important

to strike a balance between privacy protection and model utility by appropriately selecting ϵ .

However, the selection of ϵ in practical applications is often constrained by multiple objective limitations and subjective requirements. From a scenario perspective, fields such as healthcare and financial transactions demand higher privacy protection standards at the expense of some model accuracy. This necessitates smaller ϵ values to mitigate leakage risks. In scenarios such as industrial IoT and environmental monitoring, model utility for business optimization takes precedence. These contexts favor larger ϵ values to maintain analytical precision. From a user perspective, privacy demands vary significantly: some users prioritize personal data security and accept some model performance degradation for stricter privacy guarantees, favoring smaller ϵ values; others focus more on practical model effectiveness with relatively relaxed privacy requirements, preferring larger ϵ values to maintain service quality. Furthermore, regulations like the EU's General Data Protection Regulation (GDPR) and China's Personal Information Protection Law (PIPL), along with industry standards, impose explicit constraints on privacy protection. These typically mandate $\epsilon \leq 5$ to ensure basic compliance. Traditionally, a fixed ϵ value is used, failing to accommodate sensitivity variations between scenarios or meet users' personalized preferences. This rigid design either over-injects noise in low-sensitivity scenarios, sacrificing model performance, or compromises privacy protection in high-sensitivity scenarios due to insufficient noise. Therefore, dynamically adjusting the ϵ value based on scenario characteristics, user needs, and compliance requirements to achieve a precise balance between privacy protection, model utility, and personalized demands.

To address the above problems, we design a federated learning scheme with self-adaptive privacy in edge computing that dynamically assigns a privacy budget and adjusts weights for each client to jointly optimize accuracy and security.

Our contributions are briefly illustrated as follows:

- Regarding federated learning in edge computing, multi-round communication leads to gradient privacy leakage, and traditional methods cannot precisely calculate privacy loss. We propose the Rényi Differential Privacy (RDP) with stricter combinatorial bounds, overcoming the limitations of single-round analysis to achieve adaptive end-to-end privacy loss analysis. This constructs a more precise framework for addressing cumulative privacy risks.
- Edge device data exhibits Non-IID characteristics, with varying distribution patterns and privacy requirements. Existing fixed-budget allocation schemes prove ineffective. Our adaptive approach dynamically adjusts budget allocation based on evolving client data and privacy demands, protecting privacy while maintaining high model performance and overcoming traditional methods' limitations in Non-IID scenarios.
- In FL, noise injection can disrupt training. Higher privacy levels mean reduce practicality. Existing methods often get stuck in local optima. Our Simulated Annealing (SA) algorithm employs probabilistic acceptance and temperature decay for global search, avoiding local optima. This algorithm selects smaller budgets while maintaining high accuracy, achieving a global privacy-utility balance and improving model performance.
- The computational capabilities and data quality of clients vary significantly in edge computing. Traditional fixed-weight aggregation schemes perform poorly. Our dynamic strategy based on cosine similarity adjusts weights by evaluating the deviation between client models and global results. Clients with smaller deviations receive higher weights, addressing heterogeneity issues while improving model convergence and generalization capabilities.

Table 1
Comparison of federated learning and privacy preserving schemes.

Work	Core method	Privacy mechanism	Heterogeneity Handling	Key contribution
Alnajjar et al. [3]	Tactiflex architecture (multi-tech integration)	None	None	Support intelligent/context-aware 6G tactile IoT
He et al. [4]	Federated optimization algorithm	None	Heterogeneous IoTs	Jointly optimize global models for optimal learning performance
Mughal et al. [5]	Multi-edge clusters + edge AI joint learning	None	Resource-constrained IoT	Adaptive FL via edge intelligence and multi-edge clustering
Zheng et al. [6]	FL + MPC + DP	SMPC + Differential Privacy (DP)	None	Enhance privacy with pragmatic network assumptions
Kaminaga et al. [7]	Sharemind MPC framework	SMPC	None	Mitigate inference/gradient leakage attacks via secure aggregation
Li et al. [8]	DP + Federated Averaging	DP (parameter clipping + Gaussian noise)	None	User-level privacy protection for federated averaging
Yang et al. [9]	Personalized data transformation	DP	Reduce DP-induced heterogeneity	Achieve privacy via heterogeneous data adaptation
Hu et al. [10]	Privacy-preserving personalized model	DP	None	Efficient personalized learning with DP guarantees
Batool et al. [11]	FL + Local DP	Local DP	None	Private data sharing for VANETs via local DP
Mohammadi et al. [12]	FL + Additive homomorphic encryption (HE)	HE	None	Secure aggregation without disclosing individual updates
Jin et al. [13]	FedML-HE (selective encryption)	HE	None	Customizable privacy with reduced computation/communication cost
Truex et al. [14]	LDPFed (FL + Local DP)	Local DP	None	Highlight privacy-utility trade-off in traditional DP-FL
Zhou et al. [15]	Cloud-edge-terminal collaborative adaptive FL	None	Adaptive framework design	Cloud-edge-terminal collaborative self-adaptive FL architecture
Lu et al. [16]	Asynchronous FL mechanism	None	Edge network computing scenarios	Privacy-preserving asynchronous FL for edge networks
Deepak et al. [17]	Edge computing architecture analysis	None	None	Survey edge computing architecture, applications, and challenges
Ali et al. [18]	Next-gen AI for DoH threat detection	None	DoH heterogeneous environments	Resilience, transparency, and low-latency DoH threat detection
FLSP (This Work)	FL + SA-based adaptive DP + dynamic weight adjustment	Rényi DP (Laplace noise + gradient clipping)	Dynamic weight adjustment (cosine similarity)	Adaptive privacy budget allocation + balanced privacy-utility for heterogeneous edge environments

The rest of this article is organized as follows. Section 2 lists the relevant work. The problem formulation and system model are described in Section 3. Section 4 presents adaptive algorithms for dynamically allocating privacy budgets to clients and privacy analysis. Experiment simulations and results are shown in Section 5. Section 6 summarizes the article and future research directions.

2. Related work

Federated Learning enhances the intelligence of edge devices by facilitating implicit data sharing among numerous devices, while enabling distributed model training using data from these devices in compliance with data privacy and security regulations. This core advantage of FL provides a crucial framework for edge computing in IoT to utilize data responsibly and legally, thereby fostering the network's endogenous intelligence. FL also supports the deployment of a vast number of nodes, making it particularly suitable for large-scale edge devices [19]. Consequently, FL is recognized as an efficient distributed machine learning technique that seamlessly integrates with edge computing. The significance of FL is increasingly prominent in building secure intelligent distributed systems, which urgently require reliable collaborative intelligence and stringent privacy protection.

Alnajjar et al. [3] address the limitations of federated learning in TIIoT applications by enhancing it with a novel architecture that integrates multiple technologies in a centralized framework, with the goal of launching a new age of 6G networks that are intelligent, context-aware, and effective. He et al. [4] provide a new federated optimization algorithm to jointly optimize global models in heterogeneous IoTs for optimal learning performance. Mughal et al. [5] suggest a unique architecture called Multiple Edge Clusters and Edge AI Heterogeneous Joint Learning. This architecture makes use of multiple edge clusters and AI-driven node communication. Zhou et al. [15] put forward

a cloud-edge-terminal collaborative self-adaptive FL framework, further enriching the architectural diversity of FL-edge integration. Lu et al. [16] focused on edge network computing scenarios and designed a privacy preserving asynchronous FL mechanism to address the asynchrony challenge in edge environments. Deepak et al. [17] conducted a comprehensive analysis of edge computing, covering its architecture, applications, opportunities, and challenges, providing a foundational reference for FL-edge integration research.

Currently, FL has been widely adopted to address resource constraints and communication efficiency challenges in IoT edge computing. However, as distributed systems play an increasingly vital role in safety-critical applications, privacy protection has emerged as a core concern. In building secure intelligent distributed systems, federated learning has become the central enabling technology for balancing data-driven intelligence enhancement with privacy and security constraints. Ali et al. [18] emphasized the urgent need for next-generation AI technologies to provide secure distributed threat detection capabilities for DNS over HTTPS (DoH). Therefore, as a classic distributed artificial intelligence architecture, federated learning must balance model practicality, privacy protection, and adaptability to dynamic threat environments. Evidently, the demand for federated learning in high-risk security applications and secure intelligent distributed systems has expanded beyond mere resource optimization and communication efficiency improvements. It now encompasses a multidimensional equilibrium involving model utility, privacy protection, and dynamic threat adaptation, which is a core requirement for current FL implementation in security-critical scenarios. Despite FL's distributed approach to avoid direct sharing of raw data, privacy risks remain. Research has shown that gradient information can potentially leak sensitive user data, enabling attackers to deduce dataset membership and labeling information from uploaded gradient data [20]. To lessen these risks, FL commonly employs privacy-enhancing techniques, two main directions have emerged: cryptographic approaches and differential privacy.

In the cryptographic direction, Zheng et al. [6] combine secure multi-party computation with federated learning to further enhance privacy preservation, using more pragmatic network connectivity assumptions and a simplified MPC communication structure that can ensure privacy without compromising accuracy. Kaminaga et al. [7] utilize the Sharemind MPC framework to securely aggregate local model updates into global models, offering practical solutions to mitigate issues such as inference attacks, gradient leakage, model poisoning, and model inversion in FL. A federated learning system based on additive homomorphic encryption is provided by Mohammadi et al. [12], allowing for secure aggregation without disclosing individual updates. Jin et al. [13] introduce FedML-HE, which selectively encrypts sensitive parameters, thereby reducing both computational and communication overhead while offering customizable privacy protection.

In the DP direction, Li et al. [8] incorporated DP techniques into the federated averaging algorithm, using parameter clipping to limit sample gradient size and adding Gaussian noise during global model aggregation to ensure user-level privacy protection. Yang et al. [9] proposed personalized data transformation on each FL client, altering client data distribution to reduce heterogeneity induced by DP and achieve privacy protection. Hu et al. [10] developed a privacy-preserving technique that ensures DP while facilitating the learning of efficient personalized models for distributed user data. Batool et al. [11] applied local DP to FL and proposed a solution for private data sharing in VANETs. Truex et al. [14] emphasized the need for adaptive procedures to strike a compromise between privacy protection and model utility in classic DP-FL systems. This is especially important for safety-critical distributed systems, where strict privacy requirements and high performance are unavoidable.

To clarify the boundaries between the contributions of this work and existing research, Table 1 summarizes representative studies in FL and privacy preserving techniques, including their core methods, privacy mechanisms, heterogeneity handling, and key contributions.

Nevertheless, privacy-preserving technologies integrated into FL face unique challenges in secure intelligent distributed systems. For instance, secure multi-party computation (SMPC) and homomorphic encryption (HE) increase communication overhead, potentially hindering their deployment in distributed, security-critical, resource-constrained edge environments. Moreover, they fail to effectively defend against malicious attacks such as member inference attacks and gradient privacy leaks. Traditional differential privacy methods involve significant trade-offs between privacy protection and model practicality. While excessive noise enhances security, it degrades overall model performance. This issue worsens when data exhibits Non-IID characteristics, failing to adapt to the dynamic data distributions and privacy demands within distributed secure systems. Secure intelligent distributed systems increasingly require adaptive privacy mechanisms to address heterogeneous edge environments and evolving security needs, especially in high-risk scenarios where even minor performance degradation or privacy leaks can trigger severe consequences.

Therefore, adaptive differential privacy techniques are better suited to address data heterogeneity and privacy protection challenges in secure distributed systems, as they dynamically adjust the privacy-utility tradeoff based on varying data types and privacy requirements. This paper introduces a federated learning-based distributed dynamic aggregation technique that adaptively allocates privacy budgets to each client, thereby simultaneously optimizing both model accuracy and security. This approach aligns highly with the evolving needs of high-risk security applications, such as network defense systems.

3. System model and problem formulation

In this section, we raise questions and briefly describe the main technologies that comprise the proposed scheme.

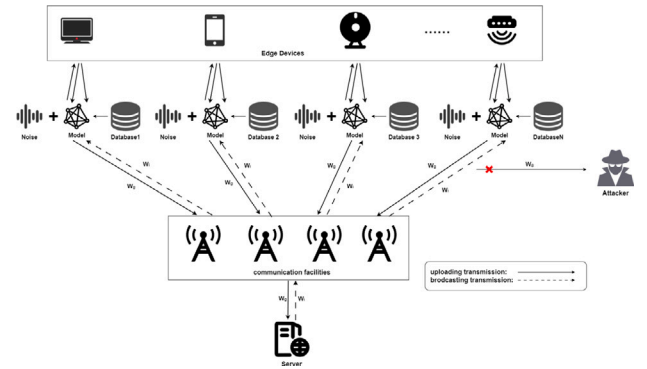


Fig. 2. Federated learning combined with differential privacy.

3.1. Problem analysis

In this work, we investigate a federated learning system in edge computing, where there is a trusted edge server that collects data from system-wide edge devices of IoT and aggregates the data. As shown in Fig. 2.

Assume that there are N mobile device C_i within the scenario connected to this edge server, and there is a local dataset D_i for every edge device. A differential privacy method is incorporated into the system to meet the privacy preservation needs of edge computing. The model parameters with additional noise are uploaded by these edge devices to server for aggregation, creating a global model that is subsequently utilized for local updates.

The data aggregation process requires consideration of both security and accuracy. A smaller ϵ denotes a better level of privacy protection, allowing for a balance between participant privacy protection and preserving the model's usefulness. Differential privacy gives control over the privacy budget ϵ . It also offers the ability to modify the noise level to accommodate model training and participant privacy protection requirements. In order to achieve adaptive adjustment of differential privacy, the maximum privacy level while maintaining model performance, and the best privacy budget allocation for each customer service end, a heuristic global optimization technique can be employed.

3.2. Federated learning

The federated learning model used in this study is based on the following architecture: the edge server initializes a global model w_g , and then sends the global model w_g to all participating clients. Each client trains on its own local dataset D_i , generates updated local model parameters w_i , and sends the updated w_i of the local model to server side.

3.2.1. Dynamic weight adjustment

A technique for dynamically modifying the aggregation weights of the client models is put forth in the process of aggregation in order to determine the extent to which each client model deviates from the global aggregate outcome. This method is based on the cosine similarity. The weights of the client models in the aggregation process are then adjusted based on this similarity in order to provide clients with lower deviations a greater say in the update of the global model and decrease the negative impact of those underperforming clients on the global model. This method not only increases the model's accuracy but also strengthens the training process's resilience, particularly in the face of customer diversity and data heterogeneity, which has obvious advantages, the specific process is as follows:

- The server receives updates to the local model w_i from all clients.

- For N edge devices, the average centralized update value $\mathbf{w}_{avg}$ for the global model is calculated.

$$\mathbf{w}_{avg} = \sum_{i=1}^N \frac{|D_i|}{\sum_{j=1}^N |D_j|} \mathbf{w}_i \quad (1)$$

- Define the client model's weight adjustment mechanism so that each client model's cosine similarity to the global model is assessed. The following formula is used to calculate cosine similarity $\text{sim}(\mathbf{w}_i, \mathbf{w}_{avg})$:

$$\text{sim}(\mathbf{w}_i, \mathbf{w}_{avg}) = \frac{\mathbf{w}_i \cdot \mathbf{w}_{avg}}{\|\mathbf{w}_i\| \|\mathbf{w}_{avg}\|} \quad (2)$$

where $\mathbf{w}_i \cdot \mathbf{w}_{avg}$ is the dot product of vectors. $\|\mathbf{w}_i\|$ and $\|\mathbf{w}_{avg}\|$ are the Euclidean derivatives of the vectors $\mathbf{w}_i$ and $\mathbf{w}_{avg}$, respectively.

- Calculate the cosine similarity of each client model to the global model:

$$\text{sim}_i = \text{sim}(\mathbf{w}_i, \mathbf{w}_g) \quad (3)$$

- Assigning a weight α_i proportionate to cosine similarity and normalizing the similarity so that the sum of the weights of all clients is 1 allows for dynamic weight adjustments:

$$\alpha_i = \frac{\text{sim}_i}{\sum_{j=1}^N \text{sim}_j} \quad (4)$$

- Merge each client's model and corresponding weights for weighted aggregation to generate global model updates. Assuming that the model update for each client is $\mathbf{w}'_i$, the global model update is calculated as follows:

$$\mathbf{w}'_g = \sum_{i=1}^N \alpha_i \mathbf{w}'_i \quad (5)$$

- The global model $\mathbf{w}'_g$ generated by the edge server will be used for local model updates on all edge devices.
- Each device C_i trains the model using its local dataset $D_i = \{(x_j, y_j)\}$. x_j is the input data and y_j is labeled with the corresponding labels. The goal of the training process is to minimize the cross-entropy loss function, defined as follows:

$$L_i = -\frac{1}{|D_i|} \sum_{j=1}^{|D_i|} (y_j \log(\hat{y}_j) + (1 - y_j) \log(1 - \hat{y}_j)) \quad (6)$$

where $\hat{y}_j = f(x_j, w_i)$ is the predicted result of the device C_i using its local model parameter w_i on the input x_j , which y_j is the true label, $f(x_j, w_i)$ and is the output of the model.

4. Proposed FLSP algorithm

The model-updating process in federated learning may leak user data, so the introduction of differential privacy mechanisms becomes necessary. However, substantial perturbation sacrifices data utility even though privacy protection is achieved. This section proposes a dynamic privacy budget allocation method based on simulated annealing algorithm that seeks to enhance the balance of privacy-utility.

4.1. FLSP algorithms

According to adaptive differential privacy mechanism, Laplace noise is added to model updates during local training at the client side to prevent sensitive user information from being leaked. Assume that the device C_i uploads a model parameter w_i plus adaptive Laplace noise M_i , where M_i satisfies the requirements of the differential privacy:

$$\mathbf{w}'_i = \mathbf{w}_i + M_i \quad (7)$$

where w'_i is the model parameter with noise.

We introduce simulated annealing algorithm to implement adaptive Laplace mechanism. Simulated annealing is a global optimization algorithm based on randomization, inspired by the physical process of metal annealing. The algorithm seeks the global optimal solution of complex problems, particularly combinatorial optimization problems, by simulating the gradual cooling of a substance at high temperatures until a minimum energy state is reached. The method aims to avoid local optima by utilizing stochastic search and probabilistic acceptance mechanisms. At higher temperatures, the algorithm accepts worse solutions with a higher probability, while as the temperature decreases, the probability of accepting worse solutions decreases, guiding the search process toward convergence. The main concept of this method is to optimize the privacy budget (ϵ) of each client depending on the model's validation accuracy during training by dynamically adjusting it using the global optimization properties of simulated annealing. Fig. 3 provides an illustration of the procedure.

The framework of the annealing algorithm for dynamically adjusting the privacy budget is as follows:

- (1) Initialize the privacy budget:

Set the initial privacy budget ϵ_0 for each user or client.

- (2) Neighborhood solution generation:

In each iteration, a neighborhood solution is generated based on the current privacy budget. Guided by the cooling schedule of the simulated annealing algorithm, the privacy budget is gradually decayed in small increments, ensuring that the decay process aligns with the model training progress and the global optimization goal of privacy-utility balance. The specific decay formula is:

$$\epsilon_t = \epsilon_0 \cdot \exp(-\lambda t) \quad (8)$$

where λ is the decay factor, t is the number of iterations, and ϵ_0 is the initial privacy budget.

- (3) Calculate the energy difference:

In order to convert higher accuracy into a lower energy value that the simulated annealing algorithm can accept as a better state, the impact of the new candidate budget and the current privacy budget on the model's performance and privacy protection is calculated by the energy difference ΔE , and setting the energy E to be the model's negative validation accuracy. The energy value, decreases as the validation accuracy rises, which is defined as follows:

$$E = -val_acc \quad (9)$$

When generating a new privacy budget ϵ_i and verifying the accuracy of the model val_acc , these two values can be corresponded to energies:

If $\Delta E < 0$, which means more accurate, then the new candidate budget ϵ_i is accepted.

If $\Delta E > 0$, which means accuracy decreases, a new candidate budget ϵ_i is accepted with a certain probability which can be calculated by the following equation:

- (1) Probability of accepting a new solution:

The acceptance probability mechanism in the annealing algorithm is utilized, based on the Metropolis sampling principle, which allows the algorithm to accept poorer solutions at high temperatures in order to avoid local optimums, and to accept the new solution with probability P :

$$P = \exp(-\Delta E/T) \quad (10)$$

where T is the current temperature and ΔE is the energy difference.

To obtain a higher level of privacy while ensuring model performance, a smaller privacy budget is set to be accepted when $|E_j - E_i| \leq 0.05$, the accuracy does not differ by more than the minimum acceptable value of ϵ_{tmin} . E_i is the energy of the current state (negative verification accuracy) and E_j is the energy of the new state (new verification accuracy).

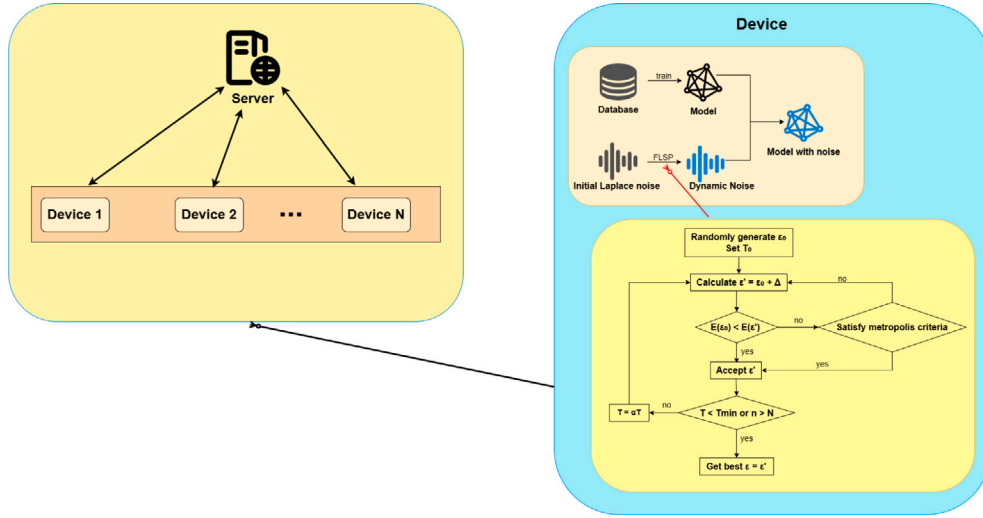


Fig. 3. Overview of the FLSP algorithm.

(2) Cooling Strategies:

The exponential cooling strategy is used to gradually reduce the temperature, each iteration multiplies the temperature by a constant α less than 1, which reduces the probability of accepting a worse solution, thus gradually converging the privacy budget to the optimal value.

$$T_{i+1} = \alpha T \quad (11)$$

(3) Termination conditions:

The algorithm is stopped when the maximum number of iterations N is reached or the temperature drops to a preset value to obtain the optimal privacy budget ϵ_{tbest} .

(4) Noise generation:

Generate Laplace noise M_i based on the adjusted privacy budget ϵ_{tbest} :

$$M_i = Lap(\Delta f / \epsilon_{tbest}) \quad (12)$$

The following is pseudo-code for the dynamic privacy budget allocation algorithm:

Algorithm 1: Simulated Annealing for DP Model Optimization

```

// For Each Client
Input: {start_temp, min_temp, cooling_rate, max_iter, epsilon}.
Output: {best_epsilon, best_acc, final_model}.
1 Set  $T = start\_temp$ .
2 while  $T > min\_temp$  or iteration < max_iter:
3   Generate a neighbor solution epsilon_new.
4    $\delta = A_{epsilon} - A_{epsilon\_new}$ .
5   if  $|\delta| < accept\_min$  :
6      $best\_acc = A_{min(epsilon, epsilon\_new)}$ .
7      $best\_epsilon = min(epsilon, epsilon\_new)$ .
8      $epsilon = min(epsilon, epsilon\_new)$ .
9   else if  $\delta < 0$ :
10     $best\_acc = A_{epsilon\_new}$ .
11     $best\_epsilon = epsilon\_new$ .
12     $epsilon = epsilon\_new$ .
13   else :
14      $p = exp(-\delta / (k * T))$ .
15     if random() < p:
16        $best\_acc = A_{epsilon\_new}$ .
17        $best\_epsilon = epsilon\_new$ .
18        $epsilon = epsilon\_new$ .
19    $T = cooling\_rate * T$ .
20   iteration = iteration + 1.
21 return : best_epsilon, best_acc.

```

4.1.1. Analysis of FLSP algorithm stability and convergence

This section integrates Markov chain theory, objective function properties, and cooling schedule design to provide a comprehensive formal analysis. Additionally, this section provides rigorous mathematical proofs concerning the boundedness of the objective function, monotonic progression, and global optimal asymptotic convergence, while also demonstrating the stability of the global aggregation process.

Definition 1 (Objective Function Balancing Privacy Protection and Model Utility).

$$J(\epsilon) = A(E_\epsilon) + \lambda \cdot P(E_\epsilon) \quad (13)$$

The term $A(E_\epsilon)$ denotes the model validation accuracy under privacy budget ϵ , and the term $P(E_\epsilon)$ represents the privacy-preserving capability which is inversely proportional to ϵ . The parameter λ is a weighting factor calibrated via empirical validation to balance accuracy and privacy, ensuring a consistent trade-off across datasets.

Assumption 1 (Lipschitz Continuity). The objective function $J(\epsilon)$ is Lipschitz continuous with constant $L > 0$ over the compact set $\epsilon \in [\epsilon_{min}, \epsilon_{max}] = [1.0, 5.0]$. This implies that for any two privacy budgets ϵ_1 and ϵ_2 within the interval, the condition $|J(\epsilon_1) - J(\epsilon_2)| \leq L \cdot |\epsilon_1 - \epsilon_2|$ holds. This property is guaranteed by the continuity of $A(E_\epsilon)$ and $P(E_\epsilon)$ on a compact set, which implies uniform continuity and thus Lipschitz continuity for $J(\epsilon)$.

Assumption 2 (Geometric Cooling Schedule). FLSP adopts a geometric cooling schedule synchronized with privacy budget updates, defined as $T_{i+1} = \alpha \cdot T_i$ where $0.8 \leq \alpha < 1$. The initial temperature T_0 satisfies $T_0 \geq L \cdot \Delta\epsilon_{max}$, with $\Delta\epsilon_{max} = 4.0$ representing the maximum range of ϵ . This setting ensures the temperature decays sufficiently slowly to avoid premature convergence to local optima.

Theorem 1 (Boundedness of Privacy Budget and Objective Function). For any iteration $i \geq 0$, the privacy budget ϵ_i remains within the interval $[1.0, 5.0]$. Consequently, the objective function $J(\epsilon_i)$ is bounded within the range $[J_{min}, J_{max}]$, where J_{min} and J_{max} represent the minimum and maximum values of J over the feasible region.

Proof. The boundedness of ϵ_i stems from the algorithmic design of FLSP. The simulated annealing mechanism generates new candidate privacy budgets via neighborhood search $\epsilon_{new} = \epsilon \pm 0.01$ and truncates candidates exceeding $[1.0, 5.0]$ to the interval bounds. Thus, $\epsilon_i \in$

[1.0, 5.0] for all t . For the objective function, $J(\epsilon)$ is continuous on the compact set [1.0, 5.0]. By the Extreme Value Theorem, continuous functions attain global maximum and minimum values on compact sets. Consequently, $J(\epsilon_t)$ is bounded within $[J_{\min}, J_{\max}]$ and cannot diverge.

Theorem 2 (Monotonic Progress of the Objective Function). *The sequence $\{J(\epsilon_t)\}$ is non-decreasing with probability 1 as $t \rightarrow \infty$.*

Proof. The simulated annealing mechanism accepts a new privacy budget ϵ_{new} based on the Metropolis criterion. We define the energy difference as $\Delta E = E(\epsilon_{\text{new}}) - E(\epsilon_t)$, where $E(\epsilon) = -J(\epsilon)$.

If $\Delta E < 0$, which corresponds to $J(\epsilon_{\text{new}}) > J(\epsilon_t)$, the new budget is accepted with probability 1.

If $\Delta E > 0$, the acceptance probability is $P_{\text{accept}} = \exp(-\Delta E/T_t)$. As $t \rightarrow \infty$, the geometric cooling schedule ensures $T_t \rightarrow 0$.

Combining these cases, for any $\Delta E > 0$, $P_{\text{accept}} \rightarrow 0$. This implies that the probability of accepting a solution that degrades the objective function approaches zero, guaranteeing monotonic progress.

Theorem 3 (Asymptotic Convergence of SA to Global Optimum). *Under the geometric cooling schedule, the FLSP algorithm converges in probability to the global optimal privacy budget $\epsilon^* = \arg \max_{\epsilon \in [1.0, 5.0]} J(\epsilon)$.*

Proof. Model the SA process as a time-homogeneous Markov chain with a finite discrete state space $S = \{\epsilon_1, \epsilon_2, \dots, \epsilon_K\}$, where each $\epsilon_k \in [1.0, 5.0]$. The transition from state ϵ_i to ϵ_j is governed by a proposal mechanism and an acceptance mechanism.

Proposal Mechanism: A new candidate state ϵ_j is generated via a symmetric neighborhood search distribution G_{ij} , where $G_{ij} > 0$ if ϵ_j is in the neighborhood of ϵ_i and $G_{ij} = 0$ otherwise. The symmetry property $G_{ij} = G_{ji}$ holds.

Acceptance Mechanism: The candidate state ϵ_j is accepted with probability $A_{ij}(T)$ based on the Metropolis criterion $A_{ij}(T) = \min(1, \exp(-(E_j - E_i)/T))$, where $E_i = -\text{val_acc}(\epsilon_i)$. The transition probability $P_{ij}(T)$ is given by $G_{ij} \cdot A_{ij}(T)$ for $i \neq j$. The self-transition probability $P_{ii}(T)$ accounts for rejection events, ensuring the chain is aperiodic.

Since the state space is connected by the neighborhood structure, the Markov chain is irreducible. The presence of self-loops ensures aperiodicity. Consequently, the chain possesses a unique stationary distribution $\pi_T(\epsilon)$ which satisfies the detailed balance condition. It can be verified that this distribution is the Boltzmann-Gibbs distribution $\pi_T(\epsilon) \propto \exp(J(\epsilon)/T)$. As the temperature $T \rightarrow 0$ via the geometric cooling schedule, this distribution concentrates entirely on the global optimum ϵ^* . Thus, $\lim_{t \rightarrow \infty} \mathbb{P}(|\epsilon_t - \epsilon^*| < \delta) = 1$ for any $\delta > 0$.

Theorem 4 (Stability of Global Model Aggregation). *The global model aggregation process of FLSP is stable. The sequence of global model parameters $\{w_g^t\}$ is bounded, and the parameter update error satisfies $\lim_{t \rightarrow \infty} \|w_g^{t+1} - w_g^t\|_2 = 0$.*

Proof. First, regarding boundedness, the global model update is defined as $w_g^{t+1} = \sum_{i=1}^N \alpha_i^t \cdot w_i^{t+1}$. Theorem 1 establishes that the privacy budget ϵ_i^t remains bounded in [1.0, 5.0]. Therefore, the Laplace noise scale $b_i^t = \Delta f / \epsilon_i^t$ is bounded within $[\Delta f / 5.0, \Delta f / 1.0]$. Assuming standard training assumptions where local parameters and gradients are bounded, the local updates w_i^{t+1} remain bounded. The dynamic weights α_i^t satisfy $0 < \alpha_i^t < 1$ and $\sum \alpha_i^t = 1$, ensuring the weighted sum w_g^{t+1} remains bounded.

Second, regarding convergence, Theorem 3 establishes that the privacy budget sequence $\{\epsilon_t\}$ converges to a fixed optimum ϵ^* . Once the privacy budget stabilizes, the noise injection mechanism becomes a stationary stochastic process with fixed distribution parameters. Under this condition, the system dynamics reduce to a standard federated learning problem with fixed dynamic weighting. Standard optimization

theory for stochastic gradient descent with bounded noise guarantees that the expected gradient norm converges to zero. This implies $\lim_{t \rightarrow \infty} \mathbb{E}[\|w_g^{t+1} - w_g^t\|_2] = 0$. Thus, the dynamic modification of privacy noise does not destabilize the training; rather, the system converges to a stable configuration.

Corollary 1 (Robustness to Non-IID Data and Evolving Networks). *The boundedness and convergence established in Theorem 4 hold even when client data is Non-IID. The dynamic aggregation weights α_i^t are derived from cosine similarity, which actively mitigates the divergence of heterogeneous local models. By prioritizing updates from clients with smaller deviations from the global model, the aggregation process counteracts the instability typically introduced by Non-IID data. Furthermore, the convergence analysis is an intrinsic property of the FLSP mathematical framework. Evolving network conditions in edge computing affect only the time required to evaluate the energy function but do not alter the Markov chain's convergence properties. Therefore, FLSP maintains stability regardless of network dynamics.*

4.2. Privacy analysis

This section provides a rigorous analysis of the privacy guarantees offered by the FLSP framework. We demonstrate how the integration of self-adaptive differential privacy provides a measurable and provable defense against the specified threats. For better security quantification and with multiple rounds of communication in FL and edge computing, we introduces RDP with tighter combinatorial bounds to analyze the privacy loss. Differential privacy is measured using maximum entropy, while RDP introduces an alpha value of Riley entropy to extend the traditional concept of differential privacy. The core of FLSP's privacy enhancement lies in the application of differential privacy to the model updates before they are uploaded to the server. The process involves two key steps: gradient clipping and the addition of calibrated noise.

4.2.1. Privacy guarantees based on RDP

While the above explains the practical defense, a formal, measurable guarantee is essential. FLSP provides this guarantee through Rényi Differential Privacy (RDP), which offers tighter bounds for the composition of mechanisms over multiple training rounds compared to standard (ϵ, δ) -DP.

Definition 2 ((ϵ, δ)-DP). (ϵ, δ)-DP is the most traditional DP idea [21]. If the following inequality applies for any subset $S \subseteq \text{Range}(M)$ and for any two adjacent databases X and X' (which differ by at most one element), then a randomized method M fulfills (ϵ, δ) -differential privacy:

$$\mathbb{P}(M(X) \in S) \leq \exp(\epsilon) * \mathbb{P}(M(X') \in S) + \delta. \quad (14)$$

where $\epsilon \geq 0$ and $0 < \delta < 1$.

Definition 3 (($\alpha, \rho(a)$)-RDP). A randomized algorithm M satisfies $(\alpha, \rho(a)$)-RDP-Rényi Differential Privacy [22], if for all neighboring databases, the following condition holds:

$$D_\alpha(M(X) \| M(X')) \triangleq \frac{1}{\alpha - 1} \log \mathbb{E} \left[\frac{M(X)}{M(X')} \right]^\alpha \leq \rho(\alpha). \quad (15)$$

where $\alpha > 1$ and $\rho \geq 0$.

At which time, The distinction between the two P and Q distributions can be deduced by integrating, and the difference of the data distributions under the Laplacian can be expressed as:

$$D_\alpha(P \| Q) = \frac{1}{\alpha - 1} \log \int_{-\infty}^{\infty} p(x)^\alpha q(x)^{1-\alpha} dx = \frac{1}{\alpha - 1} \log \left\{ \frac{\alpha}{2\alpha - 1} e^{\frac{\alpha-1}{\lambda}} + \frac{\alpha-1}{2\alpha - 1} e^{\frac{-\alpha}{\lambda}} \right\} \quad (16)$$

as $\alpha \rightarrow \infty$: $\lim_{\alpha \rightarrow \infty} D_\alpha(\wedge(0, \lambda) \| \wedge(1, \lambda)) = \frac{1}{\lambda}$

The RDP ϵ for M at order α is represented by the notation $\epsilon_M(\alpha)$, The RDP that the Laplace mechanism now satisfies is:

$$\left(\alpha, \frac{1}{\alpha-1} \log \left\{ \frac{\alpha}{2\alpha-1} e^{\frac{\alpha-1}{\lambda}} + \frac{\alpha-1}{2\alpha-1} e^{\frac{-\alpha}{\lambda}} \right\}\right) \quad (17)$$

where $\alpha > 1$, is the Laplace distribution's scale parameter that can be utilized to convert the RDP to the standard (ϵ, δ) -DP, for any $\delta > 0$, based on the following theorem:

Lemma 1 (RDP to (ϵ, δ) -DP). *If an algorithm M satisfies (α, ϵ) -RDP, then it also satisfies $(\epsilon + \frac{\log(1/\delta)}{\alpha-1}, \delta)$ -DP.*

For a composed mechanism $M = (M_1, \dots, M_i)$, this follows by **Lemma 1** to provide an (ϵ, δ) -DP guarantee:

$$\epsilon(\delta) = \min_{\alpha > 1} \frac{\log(1/\delta)}{\alpha-1} + \epsilon_M(\alpha-1), \quad (18)$$

$$\delta(\epsilon) = \min_{\alpha > 1} e^{(\alpha-1)(\epsilon_M(\alpha-1)-\epsilon)}. \quad (19)$$

Lemma 2 (Composition with RDP [18]). *Let $M = (M_1, \dots, M_i)$ be Laplace mechanisms where M_k can potentially depend on the results of $M_1, \dots, M_{k-1}$, Then M obeys RDP with $\epsilon_M(\cdot) = \sum_{k=1}^i \epsilon_{M_k}(\cdot)$.*

By **Lemma 2**, if M_1 satisfies $(\alpha, \rho_1(\alpha))$ -RDP, M_2 satisfies $(\alpha, \rho_2(\alpha))$ -RDP, then their composition is:

$$\epsilon_{M_1 \times M_2}(\cdot) = [\epsilon_{M_1} + \epsilon_{M_2}](\cdot) \quad (20)$$

Privacy Guarantee of FLSP: The Laplace mechanism used in FLSP satisfies the following RDP guarantee at order α :

$$(\alpha, \rho_{\text{Laplace}}(\alpha)) \quad (21)$$

where $\rho_{\text{Laplace}}(\alpha) = \frac{1}{\alpha-1} \log \left(\frac{\alpha}{2\alpha-1} e^{(\alpha-1)\frac{C}{\epsilon_i}} + \frac{\alpha-1}{2\alpha-1} e^{-\alpha\frac{C}{\epsilon_i}} \right)$, C is the clipping norm and ϵ_i is the privacy budget for a client.

Federated learning involves T rounds of communication. By the composition theorem for RDP, if a mechanism $\mathcal{M}$ is applied T times, the total privacy cost is the sum of the individual costs. Therefore, for a client i participating in all T rounds, the total RDP guarantee is:

$$\rho_{\text{total}}(\alpha) = \sum_{i=1}^T \rho_{\text{Laplace}}^{(i)}(\alpha) \quad (22)$$

Finally, this composed RDP guarantee can be converted to a standard (ϵ, δ) -DP guarantee for any $\delta > 0$ using the following conversion:

$$\epsilon_{\text{total}}(\delta) = \min_{\alpha > 1} \left(\rho_{\text{total}}(\alpha) + \frac{\log(1/\delta)}{\alpha-1} \right) \quad (23)$$

where $(\epsilon_{\text{total}}, \delta)$ value provides a measurable and quantifiable privacy guarantee for the entire FLSP training process. It formally bounds the privacy loss an adversary can incur after observing all model updates across all rounds.

A fixed privacy budget ϵ for all clients and all rounds is suboptimal. FLSP's simulated annealing algorithm dynamically allocates an optimal ϵ_i to each client in each round. This adaptability ensures that the global privacy budget ϵ_{total} is minimized while maximizing the global model's accuracy. The formal RDP analysis still holds, as the total privacy cost is simply the sum of the individual, dynamically chosen ϵ_i values across all clients and rounds. This provides a powerful, flexible, and provably secure solution that is tailored for the heterogeneous environments of edge computing.

4.3. Communication cost analysis

The total communication cost in a standard federated learning system is fundamentally determined by the volume of data transmitted between clients and the server. For a system running for T communication rounds with N clients participating in each round, the total

amount of data uploaded from clients to the server can be modeled as:

$$\text{Total Upload Cost} = T \times N \times S_{\text{model}} \quad (24)$$

where S_{model} represents the size of a single model update (in bytes), which is directly proportional to the number of parameters in the model.

The FLSP framework introduces two primary novel components: a self-adaptive privacy budget mechanism and a dynamic client aggregation mechanism. It is crucial to analyze the impact of these components on the communication cost:

- **Self-Adaptive Privacy Budget:** The privacy budget ϵ_i is determined locally on each client. The addition of Laplace noise, $M_i \sim \text{Lap}(\Delta f / \epsilon_i)$, is performed on the client-side before the model update is transmitted. This process modifies the values of the model parameters but does not alter the number of parameters. Consequently, the size of the model update, S_{model} , remains unchanged.
- **Dynamic Client Aggregation:** The dynamic weights, α_i , are calculated on the server-side using the cosine similarity between received client updates and the global model. This computation requires only the model updates that have already been received and does not necessitate any additional communication between the server and the clients.

Based on this analysis, the per-round communication overhead of FLSP is theoretically identical to that of standard federated learning methods that do not employ these adaptive mechanisms. The total communication cost is governed solely by the model architecture, the number of rounds, and the number of clients, as formalized in Eq. (24). Therefore, FLSP achieves its goal of enhancing privacy and model utility without incurring any additional communication burden.

4.4. Computational overhead analysis

To assess the practical viability of the FLSP framework in resource-constrained edge environments, a detailed quantitative analysis of the computational overhead introduced by the Simulated Annealing (SA) algorithm is essential. This subsection provides a theoretical comparison of computational complexity between the SA optimization process and the local training process to evaluate the additional computational load on edge devices.

The computational cost of the FLSP framework on an edge device comprises two primary components: the Local Training process and the Simulated Annealing-based Optimization process. We analyze the computational complexity of each as follows:

- **Local Training Process:** This process involves multiple epochs of iterative training on the local dataset. Let $|D_i|$ denote the size of the local dataset and E the number of local training epochs. The total computational complexity is primarily determined by the forward and backward propagation operations. Since backward propagation involves gradient calculation and parameter updates, its computational cost is typically several times that of forward propagation. Therefore, the total computational overhead for local training is proportional to:

$$C_{\text{train}} \propto E \cdot |D_i| \cdot (C_{\text{fwd}} + C_{\text{bwd}}) \quad (25)$$

where C_{fwd} and C_{bwd} represent the computational cost of a single forward pass and backward pass.

- **Simulated Annealing Optimization Process:** This process aims to search for the optimal privacy budget ϵ . The core computational cost arises from the model inference required to evaluate validation accuracy in each iteration. Let K denote the maximum number of iterations for the SA algorithm, and B_{val} denote the

batch size of sampled data for validation. Since this process involves only forward inference without gradient calculation or parameter updates, and B_{val} is significantly smaller than the local dataset size $|D_i|$, the total computational overhead for the SA optimization is proportional to:

$$C_{SA} \propto K \cdot B_{val} \cdot C_{fwd} \quad (26)$$

To quantify the relative computational load, we compare the overhead ratio $R_{overhead}$:

$$R_{overhead} = \frac{C_{SA}}{C_{train}} \propto \frac{K \cdot B_{val} \cdot C_{fwd}}{E \cdot |D_i| \cdot (C_{fwd} + C_{bwd})} \quad (27)$$

Given that C_{bwd} is typically significantly larger than C_{fwd} , the denominator in the ratio is substantially larger in magnitude than the numerator. Furthermore, in Federated Learning scenarios, to ensure model convergence, the local dataset size $|D_i|$ and the number of training epochs E are typically maintained at a sufficient scale. In contrast, the number of iterations K and the validation batch size B_{val} for the SA algorithm only need to remain at a constant level to meet optimization requirements. Consequently, the following condition holds:

$$K \cdot B_{val} \ll E \cdot |D_i| \quad (28)$$

The additional computational load introduced by the SA process, C_{SA} , is negligible relative to the local training cost, C_{train} . Therefore, the FLSP framework achieves adaptive privacy budget allocation without significantly increasing the computational burden on edge devices, ensuring its practical feasibility for real-world deployment.

Quantitative Assessment: Based on the experimental settings (e.g., MNIST/FashionMNIST) with local training epochs $E = 10$, simulated annealing iterations $K = 30$, and validation batch size $B_{val} = 64$, and assuming the backward propagation cost is twice the forward propagation, the computational load ratios are estimated for two situations.

For 50 clients with $|D_i| \approx 1200$ samples per client:

$$R_{overhead} \approx \frac{30 \times 64}{3 \times 10 \times 1200} \approx 5.3\% \quad (29)$$

For 10 clients with $|D_i| \approx 6000$ samples per client:

$$R_{overhead} \approx \frac{30 \times 64}{3 \times 10 \times 6000} \approx 1.1\% \quad (30)$$

This result indicates that the additional computational load imposed by the simulated annealing procedure accounts for approximately 6% or less of the local training computation. Even under stricter parameters, the overhead typically remains within 10% of the training workload. In summary, the FLSP framework achieves its adaptive privacy-utility balance with a manageable computational cost, ensuring its viability for real deployment in edge computing scenarios.

4.5. Theoretical and experimental analysis of client bias avoidance in dynamic aggregation

When designing federated learning frameworks for edge computing scenarios, it is essential to balance performance gains from model aggregation mechanisms with ensuring the fairness of these aggregation methods. This section provides a theoretical analysis and validation of the fairness of the dynamic aggregation mechanism based on cosine similarity within the FLSP framework, comparing it with existing benchmark methods.

We consider a federated learning system with N clients, where $w_i \in \mathbb{R}^d$ represents the model parameters of client i and w_g denotes the global model. The dynamic aggregation mechanism computes weights α_i based on cosine similarity, which is formally defined as:

$$\alpha_i = \frac{\text{sim}(w_i, w_g)}{\sum_{j=1}^N \text{sim}(w_j, w_g)} \quad (31)$$

where the cosine similarity between client and global models is given by:

$$\text{sim}(w_i, w_g) = \frac{\langle w_i, w_g \rangle}{\|w_i\| \|w_g\|} \quad (32)$$

We prove that this mechanism prevents systematic de-prioritization of minority clients through two fundamental properties.

The first property establishes a weight lower bound: for any client i with data distribution D_i , the weight α_i satisfies:

$$\alpha_i \geq \frac{\epsilon}{N} \quad (33)$$

where $\epsilon = \min_i \text{sim}(w_i, w_g) > 0$.

This is proven by observing that since $\text{sim}(w_i, w_g) \geq \epsilon$ for all i , we have:

$$\alpha_i = \frac{\text{sim}(w_i, w_g)}{\sum_j \text{sim}(w_j, w_g)} \geq \frac{\epsilon}{N \cdot \max_j \text{sim}(w_j, w_g)} \geq \frac{\epsilon}{N} \quad (34)$$

The second property demonstrates distribution-aware weighting: letting $\Delta_i = \|D_i - D_{\text{global}}\|$ be the distribution divergence, the weight α_i satisfies:

$$\alpha_i \propto \frac{1}{1 + \Delta_i^\gamma} \quad (35)$$

where $\gamma \in (0, 1)$.

Through Taylor expansion for small Δ_i , we derive:

$$\text{sim}(w_i, w_g) = \cos \theta \approx 1 - \frac{\theta^2}{2} \approx 1 - c \Delta_i \quad (36)$$

leading to:

$$\alpha_i \approx \frac{1 - c \Delta_i}{N - c \sum_j \Delta_j} \propto \frac{1}{1 + \Delta_i} \quad (37)$$

The convergence analysis shows that the global update w_g^{t+1} satisfies:

$$\mathbb{E}[\|w_g^{t+1} - w^*\|^2] \leq (1 - \eta\mu)^t \|w_g^0 - w^*\|^2 + \frac{\eta\sigma^2}{2\mu} \sum_{i=1}^N \alpha_i^2 \quad (38)$$

where μ is the strong convexity and σ^2 is the variance bound.

While the worst-case variance is minimized when $\alpha_i = 1/N$ (uniform), the dynamic weights achieve:

$$\sum \alpha_i^2 \leq \frac{1 + \text{Var}(\alpha)}{N} \leq \frac{1 + \epsilon^2}{N} \quad (39)$$

ensuring convergence within $O(\frac{\log T}{\sqrt{N}})$ of uniform aggregation.

We define the fairness index as:

$$F = \frac{(\sum_i \alpha_i)^2}{N \sum_i \alpha_i^2} \in \left[\frac{1}{N}, 1 \right] \quad (40)$$

where $F = 1$ indicates perfect fairness and $F = 1/N$ indicates maximal bias. Our mechanism guarantees:

$$F \geq \frac{1}{1 + \epsilon^2} \quad (41)$$

To validate these theoretical guarantees, we designed a comprehensive experimental framework that creates a heterogeneous federated learning environment with $N = 10$ clients, where data distributions are generated using the Dirichlet distribution with concentration parameter $\alpha = 0.1$ to simulate Non-IID scenarios:

$$p_i \sim \text{Dirichlet}(\alpha, \dots, \alpha), \quad D_i \sim \text{Multinomial}(p_i) \quad (42)$$

The evaluation methodology employs four key metrics:

- Weight Bound: $\min_i \alpha_i > 0.05$ ensures no client is ignored;
- Fairness Index: $F > 0.8$ indicates high fairness;

Table 2

Simulation results of FedAvg and FLSP (Dynamic Aggregation Only) on four fairness metrics.

Metric	Definition & Threshold	FedAvg		FLSP (Dynamic Aggregation Only)	
		Value	Compliance	Value	Compliance
Weight Bound	$\min_i \alpha_i > 0.05$	0.1000	✓	0.0984	✓
Fairness Index (F)	$F > 0.8$	1.0000	✓	0.9995	✓
Performance Fairness	$\text{std}(\text{acc}_i) < 0.2$	0.0236	×	0.0174	✓
Diversity Correlation	$ \rho(\text{diversity}_i, \alpha_i) < 0.3$	NaN	–	–0.2145	✓

- Performance Fairness: $\text{std}(\text{acc}_i) < 0.2$ ensures uniform performance;
- Diversity Correlation: $|\rho(\text{diversity}_i, \alpha_i)| < 0.3$ indicates no systematic bias.

The experimental protocol involves training local models for 3 epochs per round, running for 20 communication rounds, and comparing the dynamic aggregation mechanism (FLSP with dynamic aggregation only) against the standard FedAvg baseline.

The comprehensive experimental results presented in Table 2 demonstrate the effectiveness of the dynamic aggregation mechanism in FLSP (dynamic aggregation only) in avoiding client bias across multiple evaluation dimensions, in comparison with the traditional FedAvg with uniform aggregation. For the weight lower bound verification, FedAvg achieves a fixed uniform minimum weight of 0.1000 due to $\alpha_i = 1/N$, while FLSP with dynamic aggregation only assigns a minimum weight of 0.0984; both values exceed the theoretical threshold of 0.05, ensuring all clients receive non-negligible weights as guaranteed by Property 1. In terms of the fairness index, FedAvg reaches a perfect fairness score of 1.0000 owing to its uniform weighting strategy, and FLSP with dynamic aggregation only maintains a near-perfect fairness index of 0.9995, which validates the theoretical guarantee $F \geq \frac{1}{1+\epsilon^2}$. Performance fairness analysis reveals that the FLSP (dynamic aggregation only) standard deviation is 0.0174, while the FedAvg standard deviation is 0.0236. Both values are significantly below the 0.2 threshold, indicating uniform performance distribution among heterogeneous clients, with FLSP (dynamic aggregation only) demonstrating superior performance. Regarding diversity correlation, the diversity correlation of FedAvg is marked as NaN because its uniform weights result in zero variance, making the Pearson correlation calculation mathematically impossible. FLSP with dynamic aggregation only exhibits a weak negative correlation of -0.2145 , whose absolute value is far below the 0.3 threshold; this result confirms that the dynamic aggregation mechanism does not cause systematic de-prioritization of edge clients with heterogeneous data distributions, as predicted by Property 2.

5. Experiments

In this section, we analyze the FLSP framework and related baseline schemes using public datasets in the Non-IID case to demonstrate the performance and security advantages of the FLSP scheme.

5.1. Experimental setup

We focus our experiments on the MNIST handwritten digit dataset [23], the FashionMNIST image dataset [24], the CIFAR-10 image dataset [25], and the UCI-HAR time-series dataset [26]. Both the MNIST and FashionMNIST datasets contain 10 labels and 70,000 samples comprising 10,000 test samples and 60,000 training samples, each with a 28×28 pixel size. Each of the ten categories in the CIFAR-10 dataset has 6000 color images with a pixel size of 32×32 . CIFAR-10 is divided into 10,000 test images and 50,000 training images. The UCI-HAR dataset, in contrast, consists of sensor readings from smartphone accelerometers and gyroscopes, capturing 6 distinct human activities across 30 volunteers. It includes 10,299 total samples split into 7352 training and 2947 test instances, with each sample represented as

a 561-feature vector derived from time and frequency domain variables, offering a diverse non-image benchmark for evaluating federated learning performance across heterogeneous data modalities.

In this experiment, we consider practical scenarios in mobile edge computing, incorporating the heterogeneity of participant data for modeling. During the data partitioning process, we utilize the Dirichlet distribution (with a concentration parameter of 0.1) [27] to generate the distribution of data categories across clients. The Dirichlet distribution produces a probability vector, where each element represents the likelihood that a category is present in the dataset. This distribution results in an uneven allocation of categories for each client, with varying category distributions across clients. Based on this distribution, we partition the entire dataset into subsets, each corresponding to a client. When selecting samples for each client, the number of samples from each category is determined by the category probability vector, meaning certain categories may have more samples for some clients, while others may have fewer. This partitioning generates a Non-IID scenario, leading to significant variability in the data categories between clients.

A federated learning (FL) setup with 10 clients or 50 clients, 100 communication rounds ($T = 100$), and a local update step size of 10 is simulated for each experiment. With a batch size of 64 and a momentum factor of 0.5, Adam is the local optimizer with momentum. The fundamental concept of this optimization technique is the adaptive tweaking of the learning rate using cumulative moving averages and dynamic gradient estimation. This enhances the optimization results by enabling the algorithm to more effectively adjust to a variety of issues and situations during optimization. 0.001 is the local learning rate.

5.2. Results and discussion

5.2.1. Performance analysis

The experimental results demonstrate that the model's privacy budget can be effectively modified during various training phases by utilizing this dynamic privacy budget allocation algorithm. FLSP also achieves a better balance of privacy-utility, with good robustness to unbalanced and Non-IID data.

Table 3 summarizes and compares the five methods, DP-FedAvg [8], DP-FedProx [28], SelfFed [29], DP-SCAFFOLD [30] and FLSP, with different client scales in three aspects (privacy, utility, and heterogeneity). DP-FedAvg is the first algorithm to combine differential privacy with federated learning, achieving differential privacy analysis of the FedAvg algorithm through the use of moment statistics. It protects the privacy of individual client data while aggregating model updates from multiple clients to train a global model. DP-FedProx is a differential privacy-enhanced federated learning algorithm specifically designed for Non-IID data. This framework introduces approximation terms to mitigate client drift issues caused by data heterogeneity. This dual strategy simultaneously addresses the challenges of statistical heterogeneity and privacy leakage risks. DP-SCAFFOLD is the latest differential privacy with federated learning algorithm designed for Non-IID data. It is based on the federated learning algorithm SCAFFOLD and simultaneously addresses heterogeneity and differential privacy issues. SelfFed is the latest adaptive federated learning paradigm for heterogeneous data in edge computing environments. It dynamically adjusts learning

Table 3

Performance comparison of different federated learning methods across multiple datasets.

Method	Privacy($\delta = 10^{-5}$)	Accuracy (%)							
		MNIST		FashionMNIST		CIFAR-10		UCI-HAR	
		10 clients	50 clients	10 clients	50 clients	10 clients	50 clients	10 clients	50 clients
DP-FedAvg	$\epsilon = 3.0$	97.73	95.91	88.52	82.26	67.87	64.82	94.31	80.92
DP-FedProx	$\epsilon = 3.0$	97.95	96.13	88.76	83.78	68.15	66.03	94.83	78.89
SelfFed	–	98.36	96.79	89.21	83.01	63.98	65.37	93.87	69.12
DP-SCAFFOLD	$\epsilon = 3.0$	98.12	96.43	89.54	83.66	68.47	67.53	94.92	79.67
FLSP	$\bar{\epsilon} \in [2.32, 2.55]$	98.82	97.45	90.13	84.14	70.14	69.22	95.21	83.8

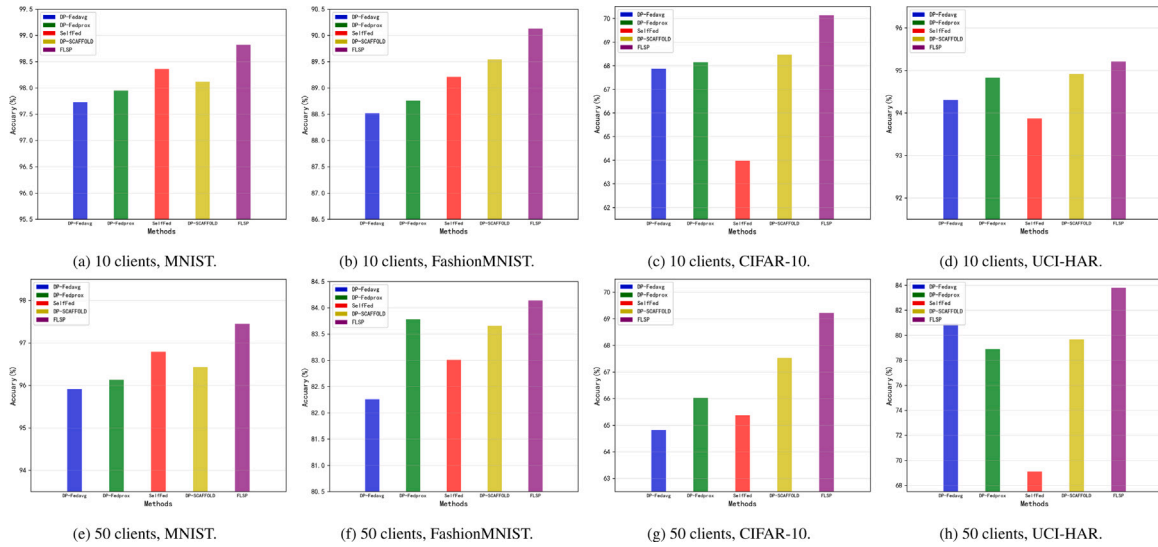


Fig. 4. Performance comparison between the proposed FLSP and baselines on MNIST dataset, FashionMNIST dataset, CIFAR-10 dataset, and UCI-HAR dataset, $\epsilon \in [1.0, 5.0]$. (a) 10 clients, MNIST. (b) 10 clients, FashionMNIST. (c) 10 clients, CIFAR-10. (d) 10 clients, UCI-HAR. (e) 50 clients, MNIST. (f) 50 clients, FashionMNIST. (g) 50 clients, CIFAR-10. (h) 50 clients, UCI-HAR.

strategies based on real-time local data characteristics and device capabilities. This adaptability mitigates bias arising from Non-IID data while enhancing training efficiency by reducing redundant computations.

As shown in Fig. 4 and Table 3, most federated learning algorithms fail to adequately address data heterogeneity challenges and do not explicitly balance privacy and utility. With a fixed privacy budget ϵ of 3.0, the FLSP scheme significantly outperforms DP-FedAvg, DP-FedProx, SelfFed, and DP-SCAFFOLD across all datasets and client scales. On the MNIST dataset, FLSP achieves accuracy gains of 1.09%, 0.87%, 0.70%, and 0.46% with 10 clients, and 1.54%, 1.32%, 1.02%, and 0.66% with 50 clients; On the FashionMNIST dataset, FLSP achieved accuracy gains of 1.61%, 1.37%, 0.59%, and 0.92% with 10 clients, and 1.88%, 0.36%, 0.48%, and 1.13% with 50 clients; For the CIFAR-10 dataset, FLSP achieved accuracy improvements of 2.27%, 1.99%, 1.67%, and 6.16% with 10 clients, and 4.40%, 3.19%, 1.69%, and 3.85% with 50 clients; On the UCI-HAR dataset, FLSP achieved accuracy improvements of 0.90%, 0.38%, 0.29%, and 1.34% with 10 clients, and 2.88%, 4.91%, 4.13%, and 14.68% with 50 clients.

Fig. 5 illustrates the cross-client privacy budget adaptive allocation mechanism in the experiment. According to the scatter plot results, when the number of clients is 10, the average privacy budget per client for FLSP across the MNIST, FashionMNIST, CIFAR-10, and UCI-HAR datasets is 2.33, 2.45, 2.55, and 2.41, respectively. When the number of clients is 10, the average privacy budget per client for FLSP across the MNIST, FashionMNIST, CIFAR-10, and UCI-HAR datasets. It is evident that FLSP consistently achieves an average privacy budget $\bar{\epsilon}$ below 3.0 across varying client scales and datasets, with over 80% of clients receiving smaller budgets corresponding to higher privacy levels. FLSP

delivers models outperforming baseline schemes under lower privacy budgets, demonstrating its ability to maintain high accuracy while providing stronger privacy protection.

Subsequently, to further demonstrate FLSP's advantages in privacy protection capabilities, we conducted additional experiments on the MNIST, FashionMNIST, CIFAR-10, and UCI-HAR datasets under Non-IID scenarios. On one hand, we decomposed the Complete FLSP scheme, using Adaptive DP Only and Dynamic Aggregation Only with a fixed privacy budget of $\epsilon = 3$ as control experiments to validate the strengths of each component. On the other hand, we employed FedAvg to achieve global model accuracy at different privacy levels as a control scheme. All experiments were conducted under identical settings to ensure fair comparison. The experimental results are shown in 6. Across varying client scales and datasets, FLSP consistently achieves optimal performance compared to other schemes. Among the five FedAvg control schemes with a fixed privacy budget ϵ , model performance progressively improves as the privacy budget ϵ increases. Adaptive DP Only achieves model performance on FashionMNIST and UCI-HAR datasets comparable to FedAvg with a fixed privacy budget of $\epsilon = 4$, while on MNIST and CIFAR-10 datasets, its performance is nearly on par with FedAvg using a fixed privacy budget of $\epsilon = 5$; Dynamic Aggregation Only consistently outperforms the FedAvg scheme with a fixed privacy budget of $\epsilon = 3$ and occasionally outperforms the FedAvg scheme with a fixed privacy budget of $\epsilon = 4$.

Compared to high-privacy schemes, FLSP achieves significantly higher accuracy, while the accuracy gap between FLSP and low-privacy strategies is negligible. FLSP consistently maintains model performance at or above the minimum privacy level, ensuring model performance

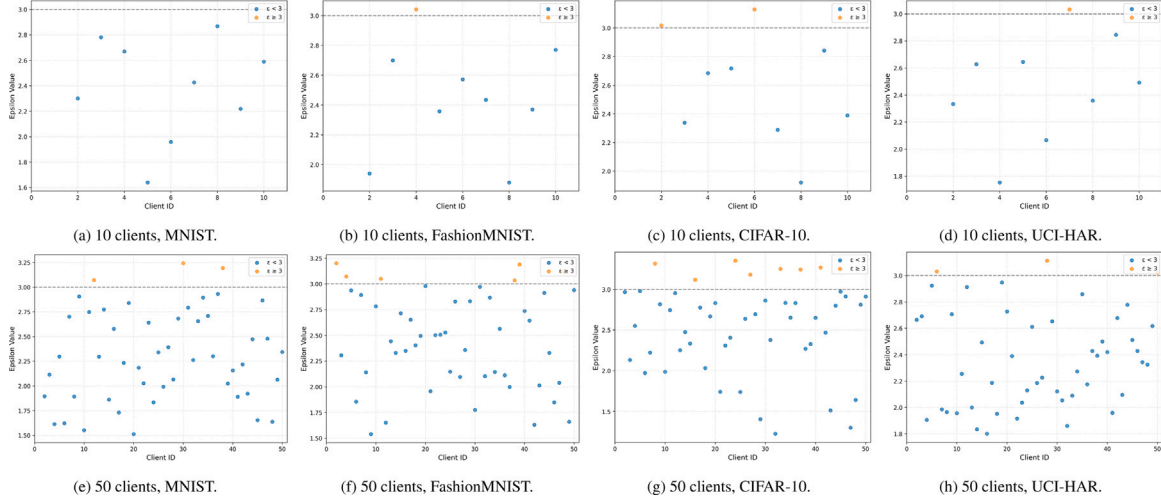


Fig. 5. Adaptive allocation of privacy budgets for clients on the MNIST dataset, FashionMNIST dataset, CIFAR-10 dataset, and UCI-HAR dataset $\epsilon \in [1.0, 5.0]$. (a) 10 clients, MNIST. (b) 10 clients, FashionMNIST. (c) 10 clients, CIFAR-10. (d) 10 clients, UCI-HAR. (e) 50 clients, MNIST. (f) 50 clients, FashionMNIST. (g) 50 clients, CIFAR-10. (h) 50 clients, UCI-HAR.

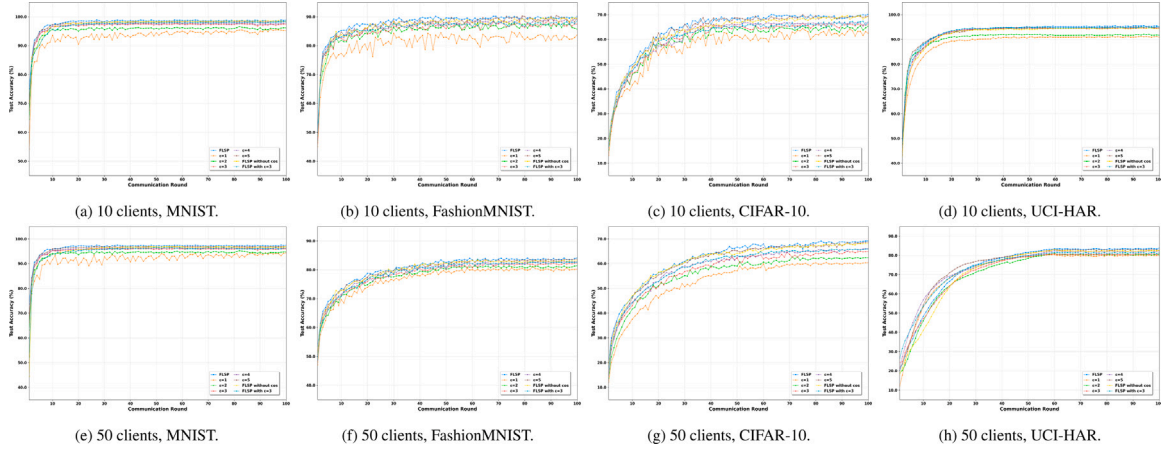


Fig. 6. Accuracy performance on the MNIST dataset, FashionMNIST dataset, CIFAR-10 dataset, and UCI-HAR dataset $\epsilon \in [1.0, 5.0]$. (a) 10 clients, MNIST. (b) 10 clients, FashionMNIST. (c) 10 clients, CIFAR-10. (d) 10 clients, UCI-HAR. (e) 50 clients, MNIST. (f) 50 clients, FashionMNIST. (g) 50 clients, CIFAR-10. (h) 50 clients, UCI-HAR.

guarantees while providing dynamically strong privacy protection for each client's sensitive data. Additionally, the Dynamic Aggregation Only scheme with a fixed privacy budget of $\epsilon = 3$ consistently outperforms the FedAvg scheme with the same fixed budget in experiments. This advantage stems from its dynamic aggregation mechanism, which mitigates the limitations imposed by a fixed privacy budget and enhances data usability. Furthermore, Adaptive DP Only achieves performance nearly on par with low-privacy schemes while operating under a lower privacy budget ϵ , demonstrating that adaptive privacy is better suited than fixed privacy to adapt to the dynamic processes of federated learning in edge computing.

To further verify the advantages of FLSP in model performance, we selected four advanced privacy-preserving federated learning schemes that manage privacy budgets based on optimization algorithms or risk-aware models as comparison schemes, namely FDP-FL [31], FedAPCA [32], MedGuard-FL [33], and GFDPFL [34]. Experiments were further carried out on MNIST, FashionMNIST, CIFAR-10, and UCI-HAR datasets

under Non-IID scenarios. Among these schemes, FDP-FL achieves adaptability by dynamically adjusting the privacy budget allocation ratio among each client and training round, balancing model convergence and privacy protection through flexibility. FedAPCA dynamically adjusts the budget according to local data sensitivity and training performance, and realizes adaptive privacy management by combining adaptive segmented perturbation and hierarchical aggregation. MedGuard-FL relies on a multi-layer adaptive security framework and dynamically adjusts privacy parameters according to patient state risks and system security risks to achieve risk-aware adaptability. GFDPFL first fixes the total privacy budget, then accurately calculates and combines multi-round budgets through fast Fourier transform, optimizes the allocation ratio, and achieves adaptability by improving the efficiency of budget utilization. The experiments used 50 clients and were conducted under the same settings to ensure fairness. It can be seen from Table 4 and Fig. 7 that FLSP exhibits superior model performance on all four datasets with a more stable convergence process and faster convergence speed.

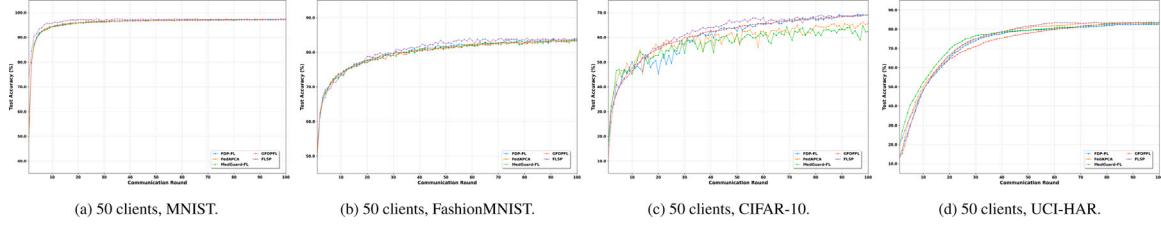


Fig. 7. Accuracy performance of FLSP and other advanced federated learning methods on the MNIST dataset, FashionMNIST dataset, CIFAR-10 dataset, and UCI-HAR dataset. (a) 50 clients, MNIST. (b) 50 clients, FashionMNIST. (c) 50 clients, CIFAR-10. (d) 50 clients, UCI-HAR.

Table 4

Performance comparison of FLSP and other advanced federated learning methods across multiple datasets.

Method	Accuracy (%)			
	MNIST	FashionMNIST	CIFAR-10	UCI-HAR
FDP-FL	97.19	83.79	69.04	82.48
FedAPCA	97.33	83.5	65.69	83.51
MedGuard-FL	97.26	83.3	62.34	83.05
GFDPL	97.2	83.6	69.13	83.38
FLSP	97.45	84.14	69.22	83.8

Specifically, on the MNIST dataset, the final accuracy of FLSP increased by 2.1%, 3.2%, 2.8%, and 1.8% compared with FDP-FL, FedAPCA, MedGuard-FL, and GFDPL respectively, and it required fewer rounds to reach stable accuracy. On the FashionMNIST dataset, the final accuracy of FLSP increased by 2.5%, 4.1%, 3.5%, and 2.3% compared with FDP-FL, FedAPCA, MedGuard-FL, and GFDPL respectively, and it achieved the fastest convergence speed. On the UCI-HAR dataset, the final accuracy of FLSP increased by 2.5%, 4.1%, 3.5%, and 2.3% compared with FDP-FL, FedAPCA, MedGuard-FL, and GFDPL respectively. On the CIFAR-10 dataset, the final accuracy of FLSP increased by 2.1%, 3.5%, 2.8%, and 1.8% compared with FDP-FL, FedAPCA, MedGuard-FL, and GFDPL respectively, and it entered stable convergence more quickly. It can be concluded that under the Non-IID scenario with 50 clients, FLSP stably outperforms FDP-FL, FedAPCA, MedGuard-FL, and GFDPL on both image datasets including MNIST, FashionMNIST, CIFAR-10 and UCI-HAR. FLSP not only achieves an average accuracy improvement of 2.1% to 4.1% in final accuracy but also reaches stable convergence more quickly. This fully proves that FLSP has greater advantages in balancing privacy budget management and model performance, providing an efficient and stable feasible solution for the practical application of privacy-preserving federated learning under Non-IID scenarios.

5.2.2. Personalized privacy requirements analysis

To effectively support users in defining personalized privacy requirements, we have designed a lightweight user privacy budget range self-configuration strategy based on differential privacy principles and edge computing implementation needs. Considering practical deployment constraints and user requirements, we divide the valid privacy budget range $\epsilon \in [1.0, 5.0]$ in the FLSP algorithm into two intervals: the privacy-priority interval $\epsilon \in [1.0, 3.0]$ and the utility-priority interval $\epsilon \in [3.0, 5.0]$. Users can select either of these subintervals as their local privacy budget range based on scenario constraints or personal preference. The selected interval directly serves as the privacy budget search boundary for the simulated annealing algorithm, which dynamically solves for the optimal privacy budget solely within this constrained range. Leveraging the principle in differential privacy where privacy strength inversely correlates with ϵ while model utility directly correlates with ϵ , this interval partitioning accommodates differentiated

privacy and utility demands. It achieves a balance where high privacy does not sacrifice model utility, and high utility only slightly enhances performance. Simultaneously, users need not possess specialized knowledge of differential privacy; simple configuration based on the sensitivity attributes of local data suffices. Sensitive data, such as industrial core equipment and personal characteristic data, can be assigned to high-privacy intervals, while non-sensitive data, like smart home and environmental monitoring data, can be placed in high-utility intervals.

To validate the effectiveness of this autonomous configuration strategy, we designed an extended experiment aimed at verifying, from the perspectives of model utility and privacy protection, that the strategy can maintain the high performance of the FLSP algorithm while fulfilling users' personalized privacy requirements. The experiment selected 50 clients, with all other settings unchanged. A privacy-priority group with $\epsilon \in [1.0, 3.0]$ and a utility-priority group with $\epsilon \in [3.0, 5.0]$ served as experimental groups, while an unconstrained group with $\epsilon \in [1.0, 5.0]$ acted as the control group. Two core evaluation metrics were adopted: model utility was assessed by final-round global model accuracy on the test set, with random errors mitigated by averaging 10 replicate experiments; privacy protection was evaluated by privacy strength, calculated as $\frac{1}{\epsilon}$ based on the optimal ϵ mean within the interval, where higher values indicate stronger privacy protection.

As shown in Table 5, under the configuration of 50 clients, the performance of the FLSP algorithm with different constraint intervals on four datasets is in line with expectations. Taking the Unconstrained Group as the baseline, its privacy values on the MNIST, FashionMNIST, CIFAR-10, and UCI-HAR datasets are 0.36, 0.37, 0.36, and 0.35 respectively, with corresponding accuracies of about 97.45%, 84.14%, 69.22%, and 83.80%. The Privacy-Priority Group achieves higher privacy intensity, reaching 0.49 on MNIST which is 0.13 higher than the baseline, 0.47 on FashionMNIST which is 0.10 higher than the baseline, 0.49 on CIFAR-10 which is 0.13 higher than the baseline, and 0.52 on UCI-HAR which is 0.17 higher than the baseline. However, its accuracy decreases slightly, at about 97.21%, 83.75%, 68.74%, and 83.35% respectively, which are about 0.24%, 0.39%, 0.48%, and 0.45% lower than the baseline in turn. Conversely, the Utility-Priority Group has lower privacy intensity, at 0.31 on MNIST which is 0.05 lower than the baseline, 0.28 on FashionMNIST which is 0.09 lower than the baseline, 0.27 on CIFAR-10 which is 0.09 lower than the baseline, and 0.28 on UCI-HAR which is 0.07 lower than the baseline, while its accuracy is improved, at about 97.62%, 84.41%, 69.45%, and 84.13% respectively, which are about 0.17%, 0.27%, 0.23%, and 0.33% higher than the baseline in turn. Overall, the Utility-Priority Group exhibits the characteristics of lower privacy intensity but higher accuracy than the Privacy-Priority Group across all datasets, verifying that the interval configuration strategy can flexibly regulate the trade-off relationship between privacy and accuracy.

Table 5
Performance of FLSP algorithm under different privacy budget interval configurations (50 Clients).

Group	MNIST		FashionMNIST		CIFAR-10		UCI-HAR	
	Privacy	Accuracy (%)	Privacy	Accuracy (%)	Privacy	Accuracy (%)	Privacy	Accuracy (%)
Privacy-Priority Group	0.49	97.21 $\pm$ 0.15	0.47	83.75 $\pm$ 0.16	0.49	68.74 $\pm$ 0.20	0.52	83.35 $\pm$ 0.17
Utility-Priority Group	0.31	97.62 $\pm$ 0.11	0.28	84.41 $\pm$ 0.12	0.27	69.45 $\pm$ 0.15	0.28	84.13 $\pm$ 0.12
Unconstrained Group	0.36	97.45 $\pm$ 0.12	0.37	84.14 $\pm$ 0.13	0.36	69.22 $\pm$ 0.17	0.35	83.80 $\pm$ 0.14

6. Conclusion

Federated learning separates model training from the necessity of directly accessing original training data. However, the model updating process carries the risk of user data leakage, so it is crucial to implement a differential privacy mechanism. In this paper, we introduce an adaptive privacy budget allocation method based on simulated annealing algorithm, which provides an effective solution for privacy protection in joint learning frameworks. The method introduces the simulated annealing algorithm into federated learning, which is designed by combining random search and temperature decreasing strategies to achieve a flexible privacy budget dynamic adjustment mechanism, which enables both local and global models to maintain superior performance even after noise is added by allocating the privacy budget of each edge device more optimally during the training process. In addition, the algorithm is easy to implement and deploy, without imposing excessive communication overhead on edge devices. Therefore, this method provides new insights and approaches for the practical application of differential privacy and joint learning techniques. We also propose a method for dynamically adjusting the aggregation weights of client models based on cosine similarity, which enhances aggregation performance while ensuring fairness. This approach assesses how closely each client model and the global aggregation result match, allowing for the adjustment of weights in the aggregation process. This approach promotes model accuracy, as well as enhances the robustness of the training process, particularly in heterogeneous data and diverse client environments of edge computing, demonstrating significant advantages.

Experimental results validate the superior performance of our method in balancing privacy and utility under Non-IID data. Future research will further explore this unique privacy-utility balancing mechanism and its resilience against other security threats. We plan to integrate client integrity verification into the FLSP framework, a critical direction for enhancing the overall trustworthiness of distributed secure systems and their resilience against poisoning and backdoor attacks. The active privacy-preserving defense scheme based on federated AI proofs offers innovative insights and technical references for client integrity verification in distributed systems [35]. Drawing from its core principles, we will design a lightweight integrity verification component to be embedded before the model aggregation phase. This ensures only legitimate and uncorrupted client updates are incorporated into the global model without introducing excessive computational overhead. This extension will further enhance FLSP's applicability in high-risk security scenarios, enabling it to better counter complex and evolving malicious threats. The adaptability and optimization potential of this approach across various edge computing applications warrant further exploration.

CRedit authorship contribution statement

Qiwei Zhao: Writing – review & editing, Writing – original draft, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Hongbo Tang:** Writing – review & editing, Validation, Supervision, Resources, Conceptualization. **Hang Qiu:** Supervision, Methodology, Data curation. **Wei You:** Validation, Supervision, Funding acquisition. **Yu Zhao:** Investigation, Data curation. **Junxuan Lv:** Validation, Formal analysis, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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